The ABBA Numerical-Method Family

Duplicated phase space, projection, composition, and configuration axes

1 Purpose and model boundary

The ABBA model is a family of integrators built from the same explicit A–B–B–A map on two copies of a physical state. A public class selects the order and projection recipe; configuration controls select the projection equations, nonlinear solver, state extension, and optional physical energy tracking. These choices are configurations of a model, not additional model identities.

The family has three public method classes:

- `ABBA2Midpoint` and `ABBA2Implicit`;
- `ABBA4Implicit`, with one outer projection.

Their focused theory documents live in the sibling model directories.

2 Duplicated-space ABBA map

Let $\dot{z} = f(t, z)$ with $z \in \mathbb{R}^m$. Duplicate the state as (u, v) and set $s = h/2$. One signed endpoint-time step beginning at t is

$$u_1 = u_0 + sf(t, v_0), \qquad v_1 = v_0 + sf(t, u_1), \tag{1}$$

$$v_f = v_1 + sf(t + h, u_1), \qquad u_f = u_1 + sf(t + h, v_f). \tag{2}$$

We denote this map by $\Psi_{h,t}(u_0, v_0) = (u_f, v_f)$. Its reverse is $\Psi_{-h,t+h}$, so the endpoint-time convention is essential for non-autonomous reversibility. When the split system is Hamiltonian, each line is a shear and the composition is symplectic on the duplicated space.

With

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \qquad P = \tfrac{1}{2}\begin{pmatrix} I & I \end{pmatrix}, \qquad G = \begin{pmatrix} I & -I \end{pmatrix}, \qquad N = G^T,$$

Ez embeds the physical diagonal, P takes its arithmetic mean, and $G(u, v) = u - v$ measures copy separation.

3 Projection choices

The midpoint method accepts $P\Psi_{h,t}(Ez)$ directly. The implicit methods use Hairer's symmetric projection. In the reduced formulation a multiplier $\mu \in \mathbb{R}^m$ displaces the input and corrects the output:

$$\widehat{Y} = Ez + N\mu, \qquad M = \Psi_{h,t}(\widehat{Y}), \tag{3}$$

$$r(\mu) = GM + 2\mu = 0, \qquad z^+ = P(M + N\mu). \tag{4}$$

If $J_\Psi = D\Psi_{h,t}(\widehat{Y})$, Newton uses

$$D_\mu r = GJ_\Psi N + 2I.$$

The simultaneous formulation solves for the corrected doubled output Y and μ together:

$$F(Y, \mu) = \begin{pmatrix} Y - N\mu - \Psi_{h,t}(Ez + N\mu) \\ GY \end{pmatrix} = 0.$$

Eliminating Y gives the reduced equation. Thus both formulations define the same ideal projected map when they converge to the same root. Newton uses the exact residual Jacobian; good Broyden replaces repeated Jacobian assembly by rank-one secant updates without changing the defining residual.

4 Canonical configuration space

ABBA2 and ABBA4 select different unprojected compositions and share one outer projection. Energy tracking is optional and passive. Signed accepted shear-stage traces update the passive momentum independently of the spatial root solve. The common state conventions are specified below.

5 Order by symmetric composition

The unprojected ABBA2 map is a symmetric method of order two in doubled space. Higher order is obtained by palindromic signed compositions, followed by one symmetric projection around the complete result. ABBA4 uses the triple jump (γ, δ, γ) with

$$\gamma = \frac{1}{2 - \sqrt[3]{2}}, \qquad \delta = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}, \quad 2\gamma + \delta = 1, \quad 2\gamma^3 + \delta^3 = 0.$$

Negative coefficients are intrinsic to these real compositions, so the dynamics must be defined at all internal signed times.

6 Projection placement is part of the model

Method	ABBA maps	projection policy
ABBA2Midpoint	1	arithmetic mean
ABBA2Implicit	1	one symmetric projection
ABBA4Implicit	3	one outer projection around the composition

The former three-projection fourth-order implementation has been removed. ABBA4 retains one outer projection. ABBA6 is removed. The retired single-projection ABBA4 factory is no longer an importable interface.

7 Finite solves and geometric claims

Symmetry, formal order, and symplecticity concern the ideal map obtained from the exact projection root. A finite stopping tolerance and floating-point arithmetic perturb that map. Consequently, the implementation records iterations, residual evaluations, final residual norms, tolerances, and projection multipliers. Step observers expose the accepted map so measured Jacobian defects can be distinguished from ideal algebraic properties.

8 Implementation correspondence

ABBA2Implicit and ABBA4Implicit declare explicit recipe tuples in `src/methods/palindromic/abba2_implicit.py` and `src/methods/palindromic/abba4_implicit.py`; ABBA2Midpoint

reuses the first tuple. ABBA2Implicit, ABBA4Implicit and BM4Implicit directly inherit `ProjectedPalindromicMethod`, which owns preparation, solver validation and advancement. ABBA4 is a complete composition with one exterior solve. There are no intermediate ABBA base classes.

The remaining public names and numerical defaults are unchanged. ABBA6 and its dedicated study, event and derivative adapter are removed without aliases. Internal imports use the defining modules; old import adapters have been removed. The ordinary implicit path is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Arithmetic midpoint variants use one explicit average instead of a nonlinear projection. Adjoint stages evaluate at the start of their signed substep; direct stages evaluate at its end.

Newton and Broyden share the spatial equations and convergence services. Exact particle-block tangents, passive energy and optional observation use retained shear inputs; energy is evaluated only after accepting the spatial map, with the existing factor 1/2 for physical κ. Time and momentum never enter the spatial root. The new internal family name does not enable the retired full time/momentum projection.

Current module responsibilities and diagrams are documented in `docs/models/palindromic/simulation/palindromic-simulation-architecture.md`.

9 Common integration lifecycle

The executor initializes a fresh numerical instance through `IntegrationMethod.new_run` and starts the shared integration driver. Implicit orders share a family base. The fixed controller supplies accepted intervals and independent shadow samples; the coordinator owns progress, collection and event dispatch. Only sampled states and accepted metric rows are retained by the collector.

10 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.